THE SHIFTED-COMPLEX CONSTRUCTION DESCENDS TO AN ADDITIVE FUNCTOR ON THE DERIVED CATEGORY FOR A NONZERODIVISOR

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Editorial note. The statement carries one footnote warning that the functor is not exact and pointing to an example; the footnote is part of the author's own statement inside the counted span and its reference is carried as a navigation stub.

Editorial note. The proof refers to a localization criterion and a formal property of localizations from other chapters and to two earlier results of this chapter; all five are navigation stubs with their roles recorded in the reasons.

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Allowed standard prerequisites (editorial).

- The homotopy category of a full subcategory closed under the relevant operations embeds as a full triangulated subcategory, and a localization criterion characterizes when the localized category is equivalent to the target. invoked at proof lines 27543-27556. Reduces the construction of the functor on the derived category to the single requirement that every complex admits a quasi-isomorphism from a complex in the subcategory.
- Every complex of modules admits a quasi-isomorphism from a K-flat complex of flat modules. invoked at proof lines 27560-27563. Supplies the required resolution, since flat modules are torsion free with respect to a nonzerodivisor.
- A functor on the homotopy category of the subcategory that sends quasi-isomorphisms to quasi-isomorphisms factors through the localization. invoked at proof lines 27566-27575. Produces the additive functor on the

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derived category whose value on a complex of torsion free modules is the shifted complex.

Source reference. Original remark 0GSN: sources/stacks-more-algebra/pinned-original-source.tex, source lines 27445–27445. The reference is cited in a footnote of the statement to warn that the functor does not preserve distinguished triangles; it is a pointer to an example and carries no part of the construction that is proved inside the counted span.

Source reference. Original lemma 0GSL: sources/stacks-chow/tags.txt, source lines 20936–20936. The reference is the localization criterion applied at proof lines 27550-27556 to reduce the construction to the existence of a torsion free resolution; the resolution argument that discharges it is inside the counted span.

Source reference. Original lemma 06Y4: sources/stacks-more-algebra/pinned-original-source.tex, source lines 14380–14380. The reference supplies the quasi-isomorphism from a complex of flat modules used at proof lines 27560-27563; flat modules are torsion free with respect to the nonzerodivisor, which is the observation the proof makes.

Source reference. Original lemma 0F7Q: sources/stacks-more-algebra/pinned-original-source.tex, source lines 27518–27518. The reference is applied at proof lines 27572-27573 to see that the shifted construction sends quasi-isomorphisms to quasi-isomorphisms, which is the hypothesis for factoring through the localization; the factorization argument is inside the counted span.

Source reference. Original lemma 04VG: sources/stacks-chow/tags.txt, source lines 6324–6324. The reference is the formal property of localizations that lets a functor sending the inverted arrows to isomorphisms factor through the localized category, invoked at proof lines 27573-27575; it is a pointer and the verification of its hypothesis is inside the counted span.

Lemma 0.2. *Let A be a ring and let $f \in A$ be a nonzerodivisor. There is an additive functor*[1] *$L\eta_f : D(A) \to D(A)$ such that if $M \in D(A)$ is represented by a complex $M^\bullet$ of f-torsion free A-modules, then $L\eta_f M = \eta_f M^\bullet$ and similarly for morphisms.*

Proof. Denote $\mathcal{T} \subset \mathrm{Mod}_A$ the full subcategory of f-torsion free A-modules. We have a corresponding inclusion

$$K(\mathcal{T}) \quad \subset \quad K(\mathrm{Mod}_A) = K(A)$$

of $K(\mathcal{T})$ as a full triangulated subcategory of $K(A)$. Let $S \subset \mathrm{Arrows}(K(\mathcal{T}))$ be the quasi-isomorphisms. We will apply Derived Categories, Lemma 0GSL to show that the map

$$S^{-1}K(\mathcal{T}) \longrightarrow D(A)$$

is an equivalence of triangulated categories. The lemma shows that it suffices to prove: given a complex $M^\bullet$ of A-modules, there exists a quasi-isomorphism $K^\bullet \to M^\bullet$ with $K^\bullet$ a complex of f-torsion free modules. By Lemma 06Y4 we can find a quasi-isomorphism $K^\bullet \to M^\bullet$ such that the complex $K^\bullet$ is K-flat (we won't use this) and consists of flat A-modules K^i. In particular, f is a nonzerodivisor on K^i for all i as desired.

[1] Beware that this functor isn't exact, i.e., does not transform distinguished triangles into distinguished triangles. See Example 0GSN.

With these preliminaries out of the way we can define $L\eta_f$. Namely, by the discussion at the start of this section we have already a well defined functor

$$K(\mathcal{T}) \xrightarrow{\eta_f} K(\mathcal{T}) \to K(A) \to D(A)$$

which according to Lemma 0F7Q sends quasi-isomorphisms to quasi-isomorphisms. Hence this functor factors over $S^{-1}K(\mathcal{T}) = D(A)$ by Categories, Lemma 04VG. $\square$